Find Smallest Subarray

with Given Sum

S = **7**

2	**1**	**5**	**2**	**3**

Min length: **infinity**

Window sum: **0**

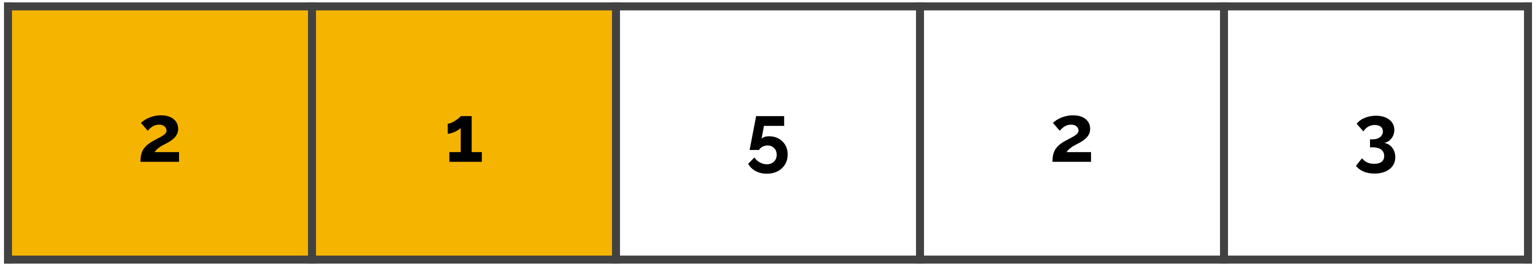
S = 7

2	1	5	2	3

Min length: **infinity**

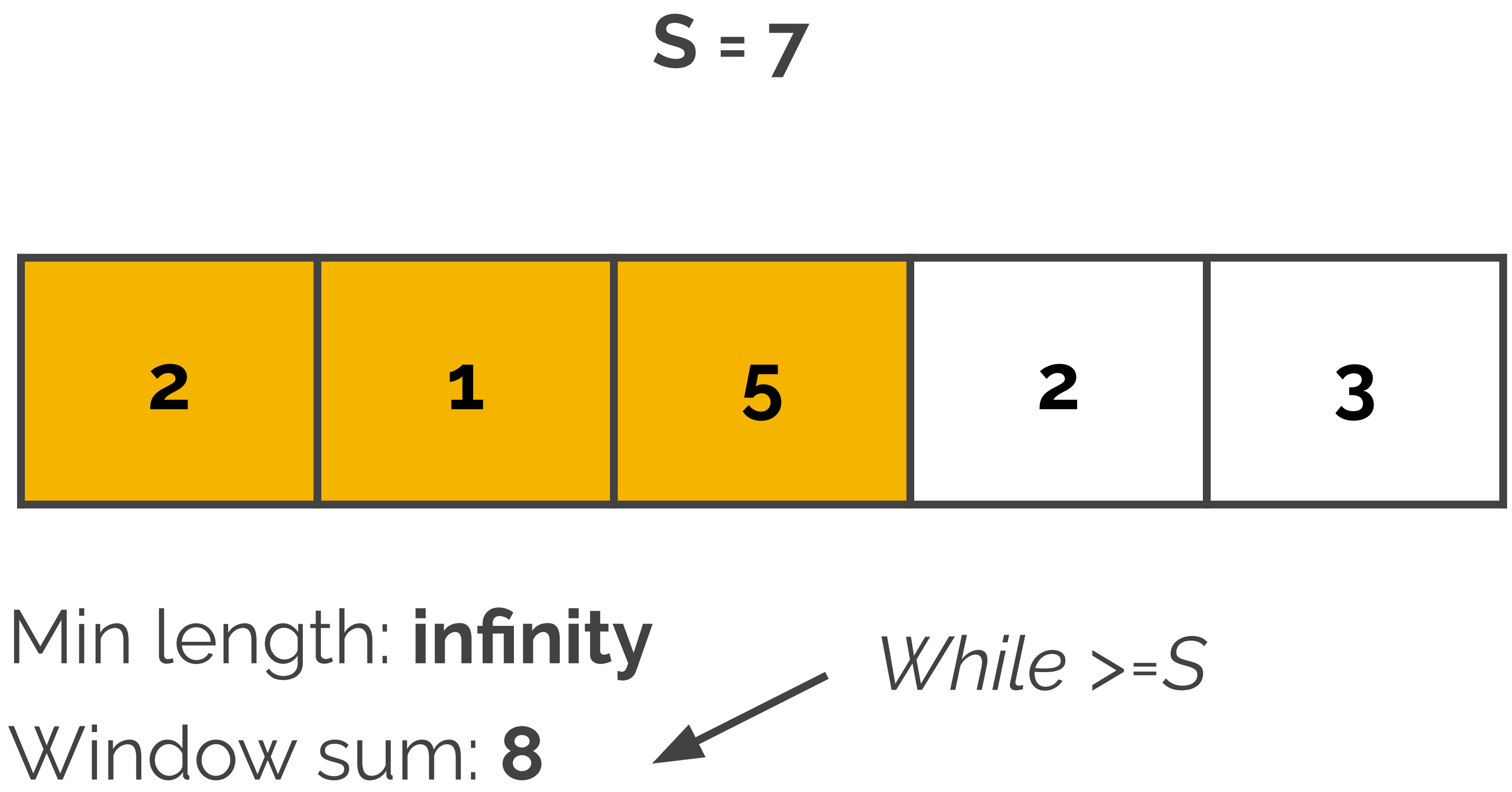
Window sum: **2**

S = 7



Min length: **infinity**

Window sum: **3**

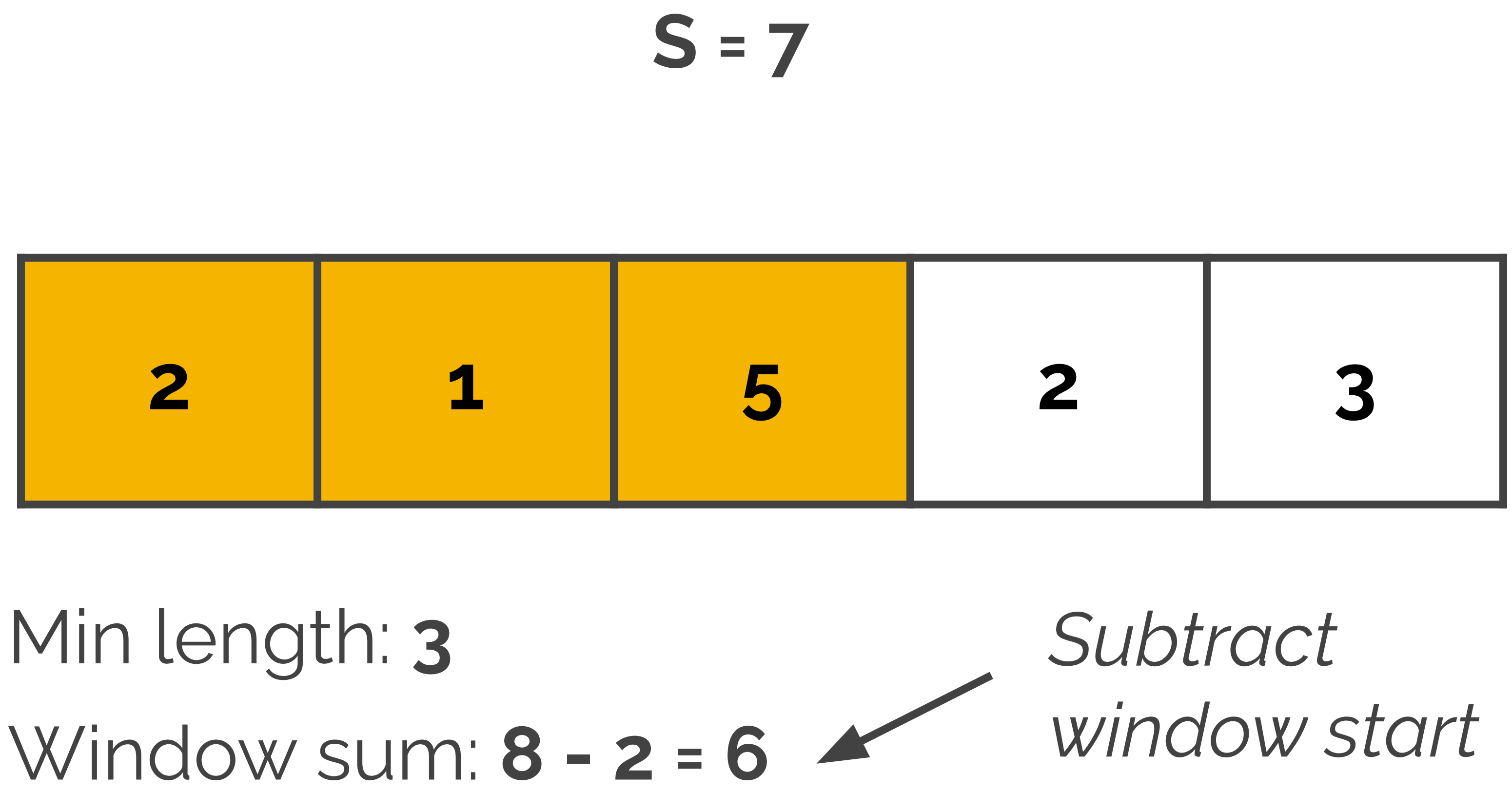

S = 7
2
1
5
2
3
Min length: infinity
Window sum: 8
While >=S

S = 7

Min length: **infinity**

Window sum: **8**

Find min of ***min length*** *and* ***current index - start index + 1***

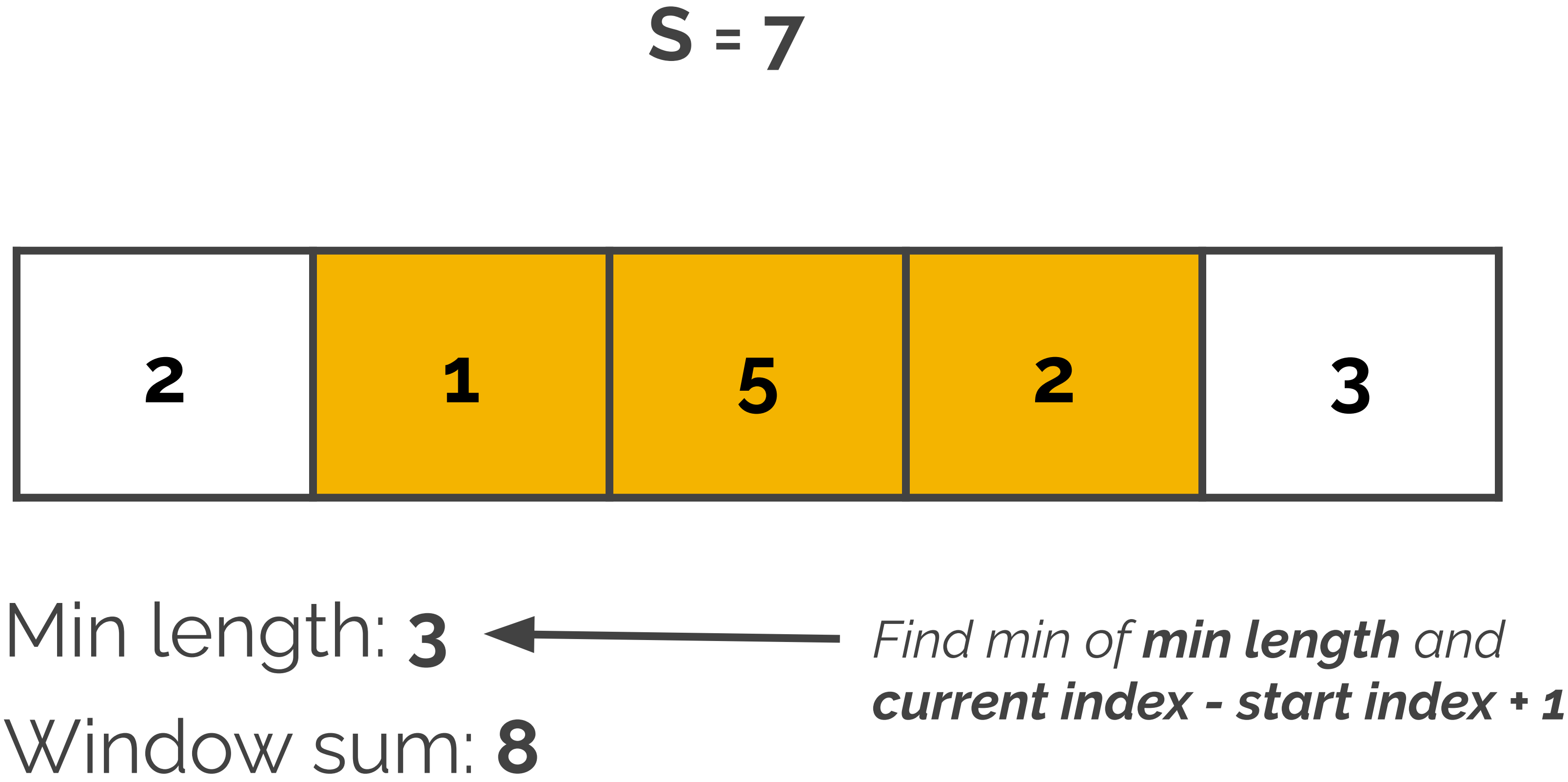

S = 7
2
1
5
2
3
Min length: 3
Window sum: 8 - 2 = 6
Subtract window start

S = 7

Min length: **3**

Window sum: **8**

While >=S

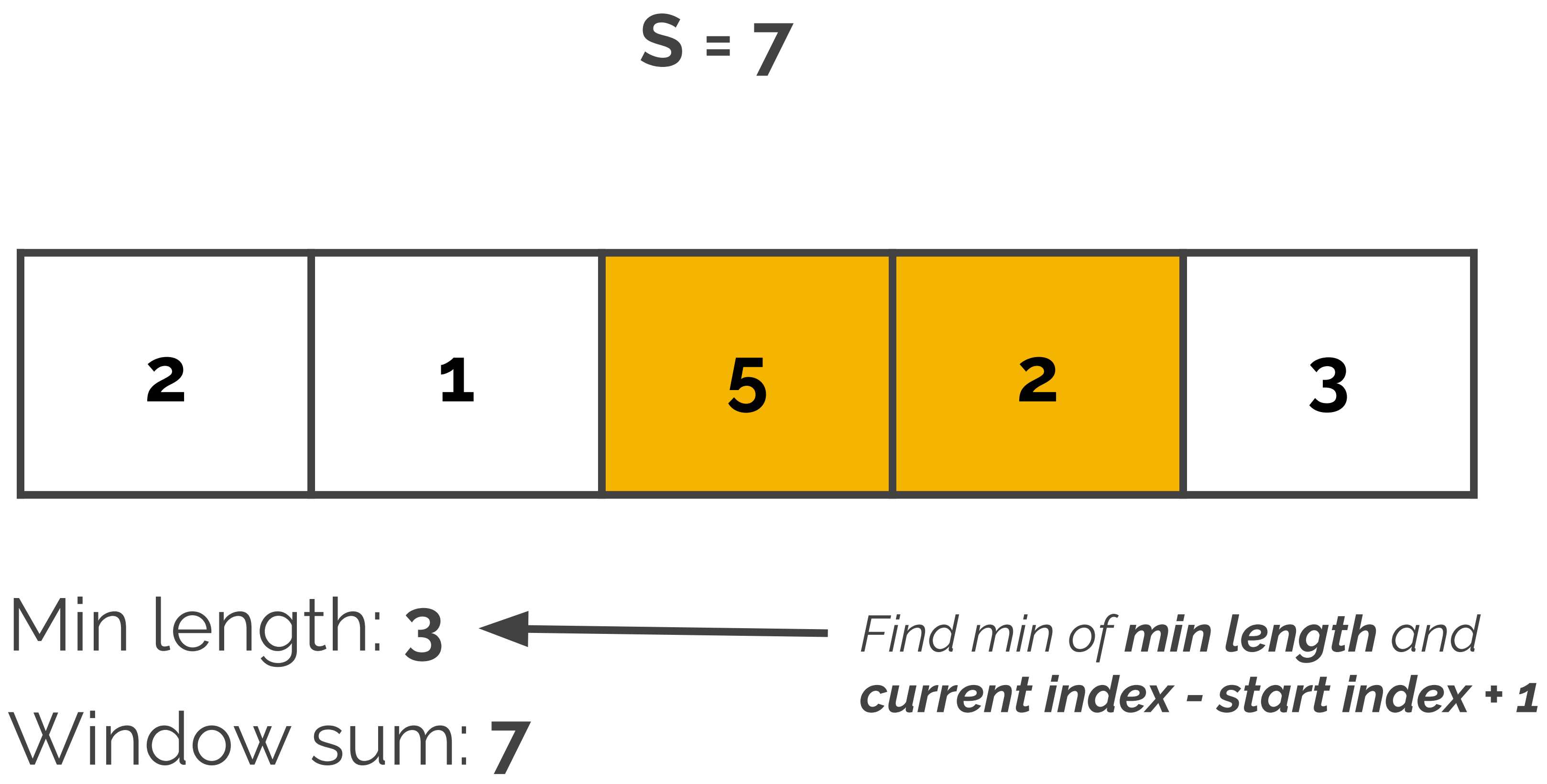

S = 7
2
1
5
2
3
Min length: 3
Window sum: 8
Find min of min length and
current index - start index + 1

S = 7

Min length: **3**

Window sum: **8 - 1 = 7**

Subtract window start


S = 7
2
1
5
2
3
Min length: 3
Window sum: 7
Find min of **min length** and
current index - start index + 1

S = 7

2	1	5	2	3

Min length: **2**

Window sum: **10**

While >=S

S = 7

Min length: **2**

Window sum: **10**

*Find min of **min length** and*
current index - start index + 1

Time complexity **O(N)**

- The outer for loop runs for all elements.
- Inner while loop processes each element only once
- Time complexity of the algorithm will be O(N+N) which is asymptotically equivalent to O(N).

Space complexity **O(1)**